1. Aerodynamic forces and coefficients:

   For any object in a fluid flow, the aerodynamic forces, e.g. lift (L), drag (D), are calculated:

$$L = \frac{1}{2}\rho v^2 A C_l,\ D = \tfrac{1}{2}\rho v^2 A C_d$$

   Where:

   $\frac{1}{2}\rho v^2$, dynamic pressure, quantifies the kinetic energy per unit volume of a moving fluid.

   ρ: Fluid density. v: Velocity of the fluid. A: Reference area.

   C_l C_d: The dimensionless lift and drag coefficients

   Typically, for a flat object, this lift coefficient is linear when small AOA, $\alpha < 10°$,

$$C_l = m \cdot \alpha$$

   where α is an angle of attack, and m is a slop constant, say 0.11 per degree.

   The drag coefficient is modeled from two parts: skin friction and pressure drag:

$$C_d = C_{d,0} + k\alpha^2$$

   For a rocket fin, $C_{d,0}$ typically ranges from 0.02-0.05, Square/un-sanded or Rounded edges, and k 0.0005-0.002.

   * Aerodynamic forces are distributed, but simplify to concentrated at a point, called Aerodynamic Center (AC)

2. Aerodynamic forces applied to a rocket

   The master Equation of total drag coefficient is simply sum up from each component (be sure they are scaled to a single reference area)

$$C_{d,\text{total}} = C_{d,\text{nose}} + C_{d,\text{body skin}} + C_{d,\text{fins}} + C_{d,\text{base}} + C_{d,\text{interference}}$$

   Center of pressure (CP) considers sideways tilting forces (Normal forces), Barrowman Equation takes a weighted average of each components' projected locations

$$CP_{\text{total}} = \frac{(C_{N,\text{nose}} \cdot X_{\text{nose}}) + (C_{N,\text{body}} \cdot X_{\text{body}}) + (C_{N,\text{fins}} \cdot X_{\text{fins}}) + \ldots}{(C_{N,\text{nose}} + C_{N,\text{body}} + C_{N,\text{fins}} + \ldots)}$$

   You may details your own components like conical shoulder (between nose and body), boat-tail tube (behind fins), booster connector, etc.

3. Stability rules

   1. Put the Center of Gravity (CG) in front of CP
   2. CG and CP cannot be too close or far away (Golden: 1-2x diameter of rocket.